

## Scramble the message!

Now that you've got your interface working, it's time to write the code that will encrypt and then decrypt your secret message.

### 8 Is it even?

You need to create a function to tell the program whether or not there's an even number of characters in your message. The function will use the modulo operator (%) to check if it can divide the number by 2 without leaving a remainder. If it can (True), then the number's even. Add this function under the code you typed in Step 2.

```
def is_even(number):
    return number % 2 == 0
```

Pass the True or False value back to the code.

This will be True if the number is even.

### 9 Get the even letters

In this step, you'll make a function that takes a message and produces a list containing all the even-numbered letters. The function uses a **for** loop with a range that goes from 0 to **len(message)**, so that it checks all the letters in the string. Add this function under the code in Step 8.

```
def get_even_letters(message):
    even_letters = []
    for counter in range(0, len(message)):
        if is_even(counter):
            even_letters.append(message[counter])
    return even_letters
```

Make a list variable to store the even letters.

Loop through every letter in the message.

If this is a letter in an even position, Python adds it to the end of the list of letters.

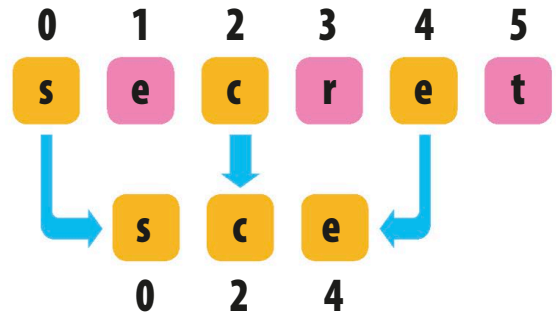
Pass the list of letters back to the code that called this function.



### EXPERT TIPS

#### Modulo operator (%)

If you put the modulo operator (%) between two numbers, Python tells you the remainder when you divide the first number by the second. So  $4 \% 2$  is 0, but  $5 \% 2$  is 1, because there's 1 left over if you divide 5 by 2. Type these examples in the shell if you want to try them out.



Don't forget to save your work.